

Cognitive and Neural Science

ITSC-6131

The Cognitive Approach

Adane Mamuye

adane.letta@aau.edu.et

March 2026

Basic Brain Processes

- Visual recognition
- Long-term memory
- Short-term memory
- Action selection
- Motor control

Approaches in Cognitive Science: The Cognitive Approach

- Starting in the 1960s a new form of psychology arrived on the scene.... known as cognitive psychology
- Cognitive psychologists placed renewed emphasis on the study of internal mental operations.
- Adopted the computer as a metaphor for mind, and described mental functioning in terms of representation and computation.... inherent in the cognitive approach is the idea of modularity
- Cognitive psychologists have studied a wide variety of mental processes, including pattern recognition, attention, memory, imagery, and problem-solving.

Background

- Is the newest and is currently its most influential.
- Behaviorists avoided the study of mental phenomena because they believed that they were too difficult to define or measure.
- In some cases behaviorists may have denied its existence, the cognitivists firmly acknowledged the existence of mental processes and focused their investigatory attentions on them.

Behaviorism focused exclusively on observable behaviors and dismissed the study of internal mental processes as unscientific.

Background

- Cognitive psychology can be seen as a counter-revolution against the behaviorist movement.
- There were three main reasons for the rapid growth of this new perspective.
 - The failure of behaviorism to account for findings in areas such as language acquisition.
 - The invention of new measuring devices to examine mental activity.
 - The rise of the computer and the widespread use of the metaphor of mind-as-computer.

The Cognitive Approach: Mind as an Information Processor

- **Cognitive psychology** is the study of knowledge representation and use in human beings.
- Concerned with understanding how people represent, process, and store information.
- Cognitive psychologists adopt the scientific method as their primary tool of investigation.
 - Hypotheses are tested by analyzing data that has been obtained from controlled experiments.

The Cognitive Approach: Mind as an Information Processor

- Cognitive psychology also supplements experimentation with modeling and computer simulation.
- A specific information-processing model of a mental ability can be run on a computer.
- Parameters of the computer model can be refined so as to provide a better fit between the computer model and the empirical data.
- Simulation can also yield insights that cause researchers to go back and design new experiments.

The Cognitive Approach: Mind as an Information Processor

- A defining characteristic of the cognitive approach is the way it represents human information processing.... often conceptualized using a **process model**.
- **Process models** are a very important part of the cognitive perspective.
 - Are a powerful conceptual tool for understanding human information processes.

Stage	Operation Type	Purpose
Encoding	Perception	Creates a mental symbol of the stimulus.
Comparison	Computation	Matches the symbols against memory stores.
Decision	Logic	Determines if a match exists (Yes/No).
Response	Motor Output	Executes the physical action.

Modularity of Mind

- An assumption of many cognitive theories is **modularity of mind**.
 - The mind is made up of innate, functionally independent modules.

“Jerry Fodor is the outspoken proponent of the modular view. In his book *The Modularity of Mind* (1983), he gives a general description of the role these putative structures play. In his view, information arriving from the outside environment passes first through a set of sensory transducers.”

- Fodor lists the many characteristics of a module as hardwired, meaning that they cannot be constructed from more basic units.

Evaluating the Modular Approach

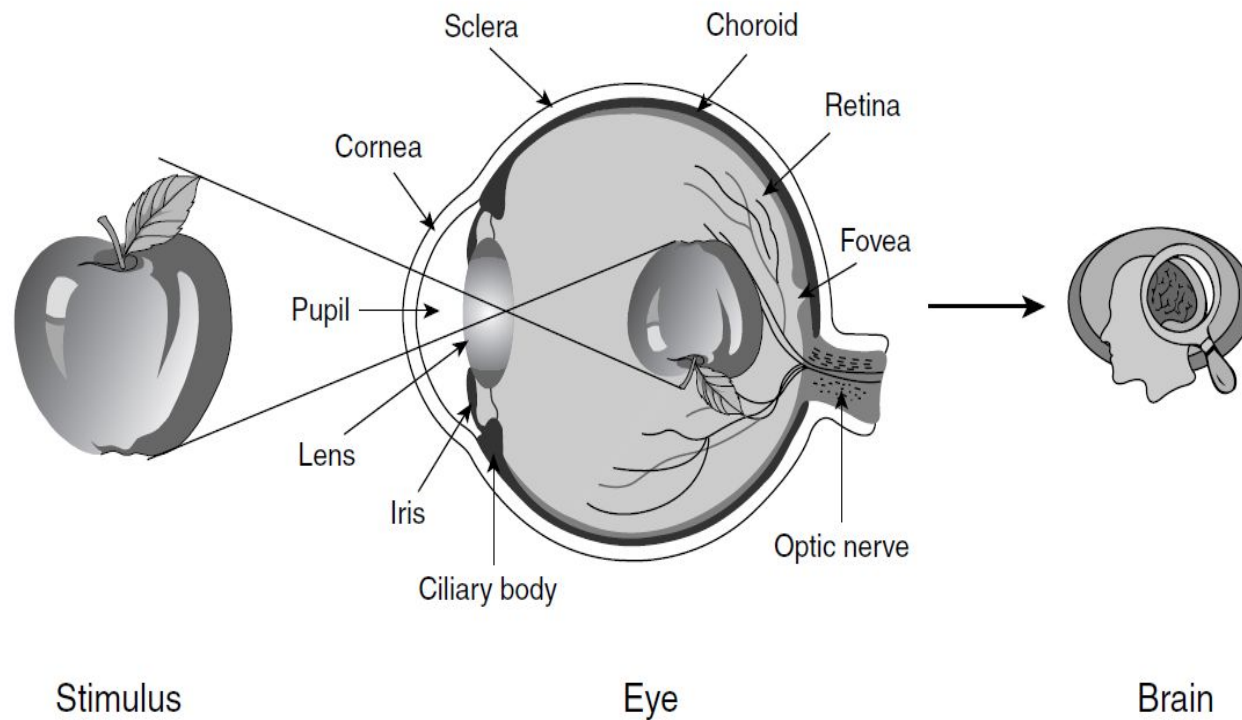
- Case studies of brain-damaged patients support the modular view
- There are many case studies of patients who have suffered damage to a particular brain area as a result of a stroke or accident.
 - Damage to another region might cause difficulties in his or her speaking or producing language.
 - Patients with developmental genetic disorders in whom one language ability has been preserved while another has not.
- Current thinking is that the brain may contain a number of processing structures that are modular, but that these may be a result of normal developmental processes.

Theories of Vision and Pattern Recognition

- **Perception** is the process by which we gather information from the outside world via the senses and interpret that information.
- Most work in perception has focused on vision and secondarily on audition.
- Vision is our most sophisticated sense and the one we rely on most.
- One of the main functions of a visual system is the recognition of patterns.
- **Pattern recognition** is the ability to identify objects in the environment.
- This process is actually quite complex and far from being entirely understood.

Theories of Vision and Pattern Recognition

- **Template Matching Theory**

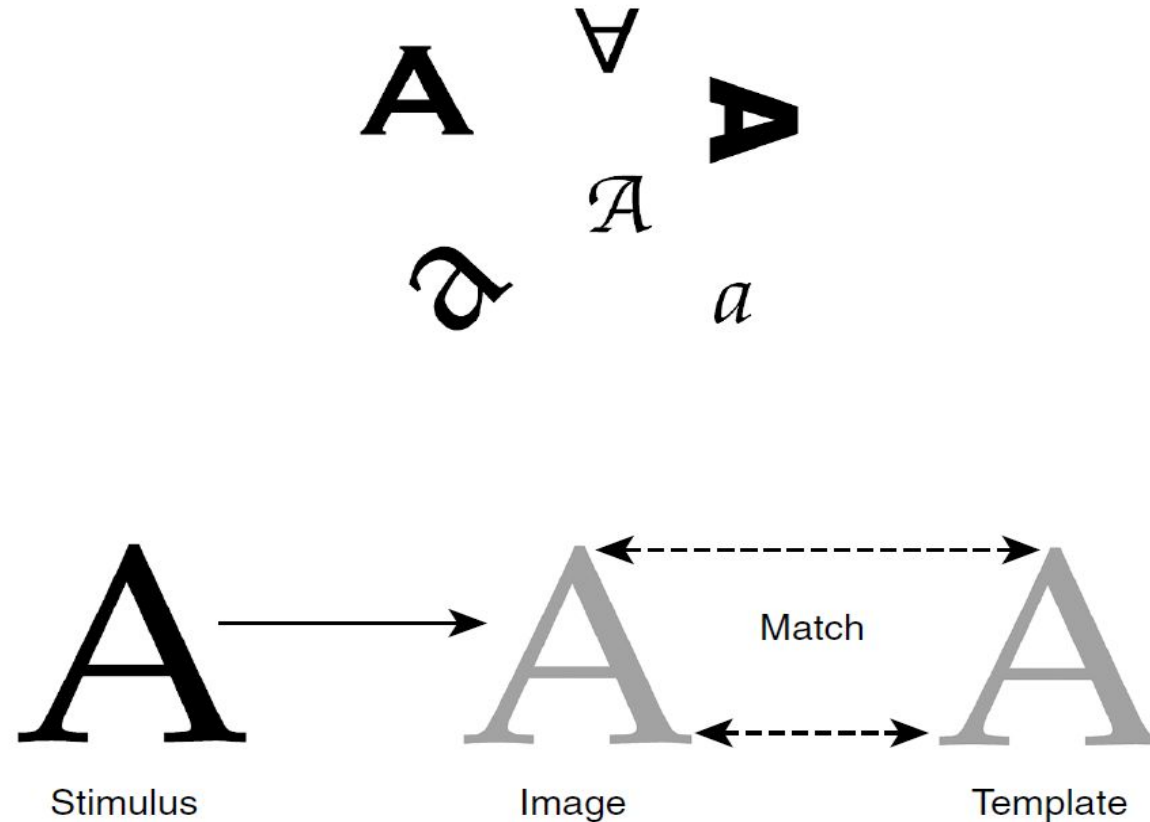


- A pattern recognition process must begin with the stimulus (the actual object in the external world).
- Light striking the stimulus is reflected and projects an inverted image of the stimulus onto the retina.
- The retina performs some preliminary processing of the image... stimulus is passed posteriorly along a set of pathways toward visual areas in the brain.
- It is in these areas where the bulk of pattern recognition and other perceptual processing takes place.

An image generated by an external stimulus is matched to an internal mental representation of the stimulus, called a **template**. The degree of overlap between the image and the template is then computed.

Theories of Vision and Pattern Recognition

Template Matching Theory



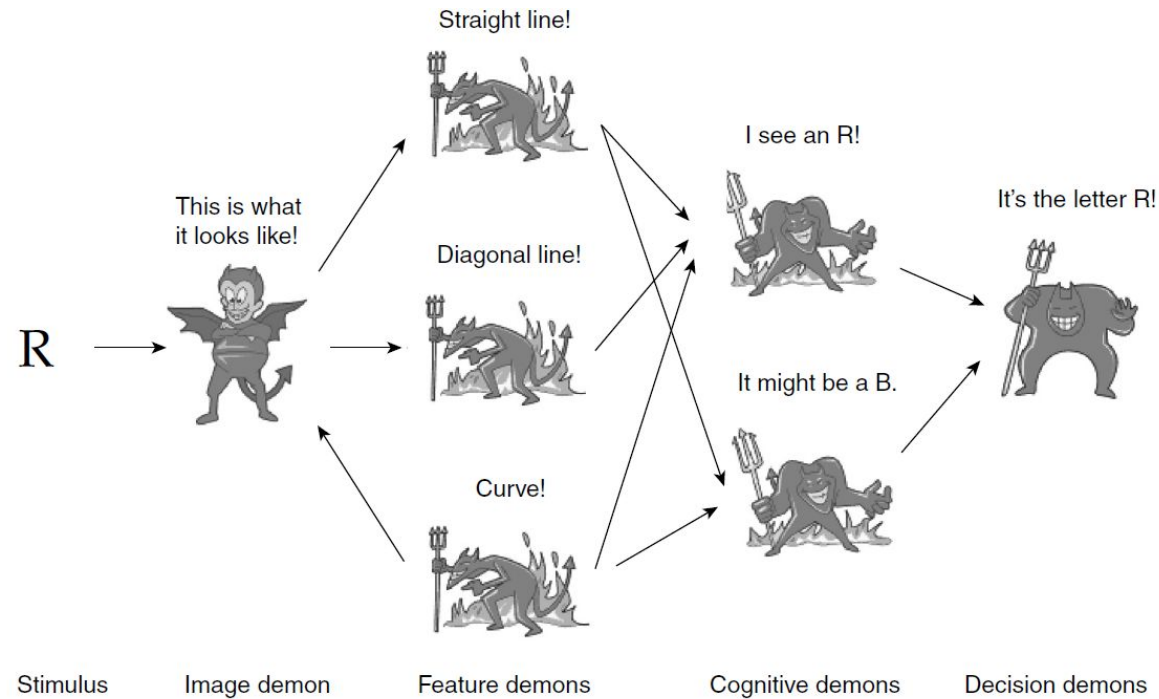
The top panel provides various depictions of the letter “A,” all recognizable. The bottom panel shows the sequence of events in the template matching model

Theories of Vision and Pattern Recognition

- **Feature Detection Theory**

- Instead of your brain looking at an object like a single photograph (Template Matching), it looks at it like a collection of building blocks.
- With feature detection models, an image of a stimulus, rather than being matched in its entirety to a template, is broken down into its component features.
- A feature is a part or subset of an object.
- The idea is that each different combination of features uniquely specifies a different object.
- The best-known feature detection model is called the **pandemonium model**

Theories of Vision and Pattern Recognition



The pandemonium model of pattern recognition. Different “demons” perform different steps in the recognition of the letter R

Theories of Vision and Pattern Recognition

- *Evaluating Feature Detection Theory*

- The **pandemonium model** of pattern recognition represents a significant improvement over template matching.
- Another reason to like the pandemonium model comes from neurophysiology.
- Evidence from this field demonstrates that neurons in the visual system act as feature detectors.
- Feature detection models are also bottom-up, or **data-driven processes**.

While this theory is robust for simple visual recognition (like reading letters), critics argue that bottom-up processing alone cannot explain everything.

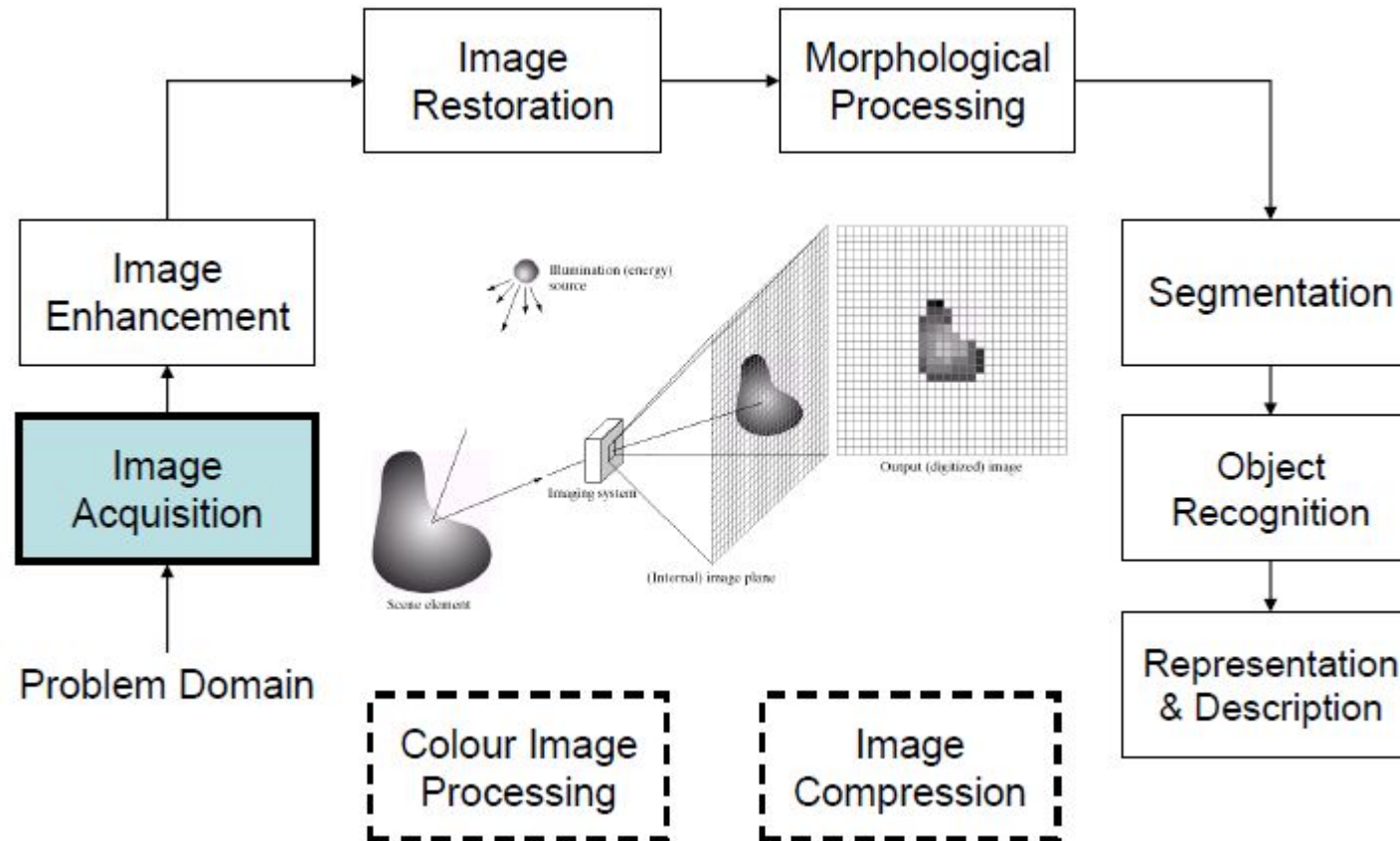
Theories of Vision and Pattern Recognition

- **A Computational Theory of Vision**

“In 1982 the renowned cognitive scientist David Marr wrote a book that described his influential theory of pattern recognition. He adopts a computational approach to vision. In this approach, the visual system is treated as a computer.”

- The “computer” receives an image of an object and, after performing several algorithmic processing steps, accomplishes recognition of the object.
- Each major processing step produces a more fully articulated version of the object—what Marr calls “sketches.”

Theories of Vision and Pattern Recognition



Theories of Attention

- **Attention** is defined as concentrated mental activity.
- Attention as a form of mental activity or energy that is distributed among alternative information sources.
- Informational sources can be stimuli from the outside world or thoughts or other forms of internal mental content.
- From a cognitive perspective attention can be considered as having two sets of opposing characteristics.

Theories of Attention

- It is selective but divisible, as well as shiftable but sustainable.
- **Selective** mean that attention can be focused onto one source of information and away from another.
- At the same time attention is **divisible**.
- Can be split or divided among several alternative information sources.

“The greater the number of sources among which attention is divided, the less attention there is to devote to any single source and the worse is the individual performance with respect to each.”

Theories of Attention

- One can selectively attend to information source A while ignoring B, then switch back to B and ignore A.
- The point is that attention can be repeatedly shifted among different sources.
- This shifting can be voluntary or involuntary
- **Distraction** is an example of the involuntary shifting of attention to some irrelevant or undesirable piece of information.

Theories of Attention

- The converse of shifting is **sustainability**.
- We can think of sustained attention as extended focus or concentration.
- There are two general classes of theories that attempt to explain attention.

Theories of Attention

- **Bottleneck theories** describe why it is that of all the information that is presented to us, only a small portion of it actually gets through.
- These theories attempt to explain the apparent bottleneck effect or the narrowing down of the information that reaches conscious awareness.
- Bottleneck theories are inherently theories of selective attention because they describe how some information is selected for processing as the rest gets discarded.
- **Capacity theories** on the other hand are essentially theories of divided attention.
 - Resources flexibly allocated to tasks based on their demands and the individual's cognitive capacity
- They conceptualize attention as a limited resource that must be spread around different informational sources.

Theories of Attention

- **Broadbent's Filter Model**

- The British researcher Donald Broadbent did some of the earliest pioneering work in cognitive psychology.
- He proposed a theory to account for why we can't follow two streams of information coming in simultaneously via the two ears.
- The dichotic listening task has been widely used to study selective attention.
- In this paradigm, a participant wearing headphones listens to two different messages being played, one over each ear.

Theories of Attention

- **Broadbent's Filter Model**

- For instance, a male voice may be reciting a list of words into the left ear as a female voice recites a different list of words into the right ear.
- The participant's task is to pay attention to, or shadow, one of the voices and to ignore the other.
- The messages or information sources presented to the participant from both ears are in this context referred to as channels.
- The participant is asked to recall what he or she has heard, for both the shadowed and unattended channels.

Theories of Attention

- Early dichotic listening studies showed that participants were fairly accurate in recalling content from the shadowed ear, but quite poor for recalling that coming from the unattended ear (Cherry, 1953)
- To Broadbent, this suggested the presence of a filter that could block out messages that were being relatively ignored, allowing only attended information to enter awareness.

Theories of Attention

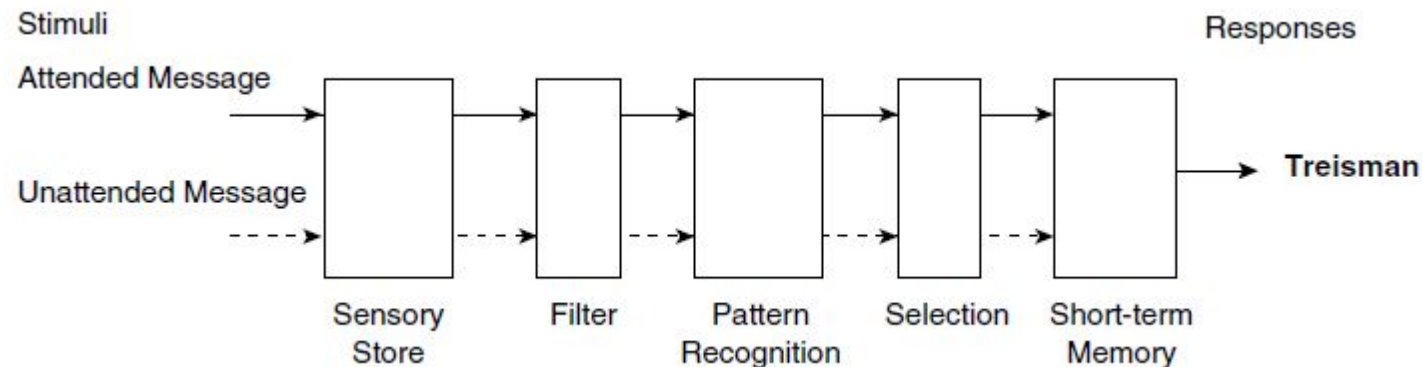
- *Evaluating the Filter Model*

- One problem is summed up in what is one of the better names for a psychological phenomenon, the so-called **cocktail party effect**
- The observed cocktail party effect and other studies have demonstrated that some information gets through the unattended channel.
- The filter does not block words of great personal relevance, such as your name and words associated with danger, for example, “fire.”

Theories of Attention

- **Treisman's Attenuation Model**

- Attenuation theory is similar to filter theory but with one exception
- Instead of completely blocking out the unattended message, the filter attenuates or weakens it.
- Message attenuation is like having water running through two faucets, where each of the faucets constitutes a different information channel.
- The faucet corresponding to the shadowed message would be open all the way, allowing water to pour through.



Theories of Attention

- Treisman explains that different words have different chances of making it through the unattended channel, due to a threshold effect.
- A **threshold** is the minimum amount of activation required to produce conscious awareness of a stimulus.
- Stimuli with associated low thresholds easily make their way into awareness. Those with associated high thresholds do not.

Theories of Attention

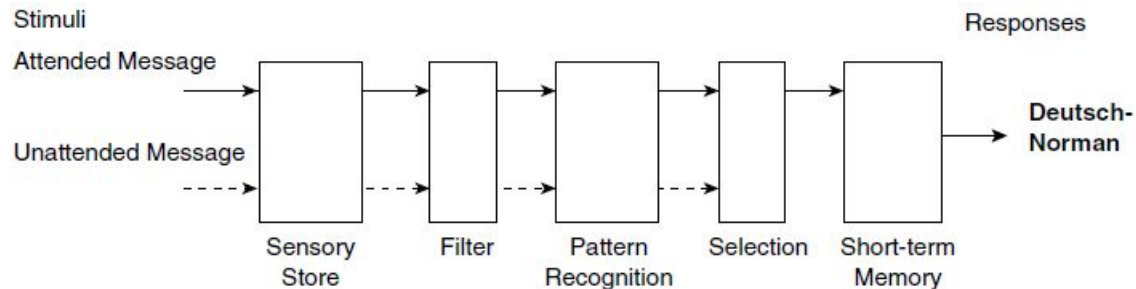
- **The Deutsch-Norman Memory Selection Model**

- The cocktail party effect shows that meaning, or semantics, is another criterion according to which information is selected.
- Deutsch and Deutsch (1963) and Norman (1968) therefore proposed a second selection process, one that is based on semantic characteristics.
- This selection happens later on in processing.
- Their model is thus an example of a late selection model.

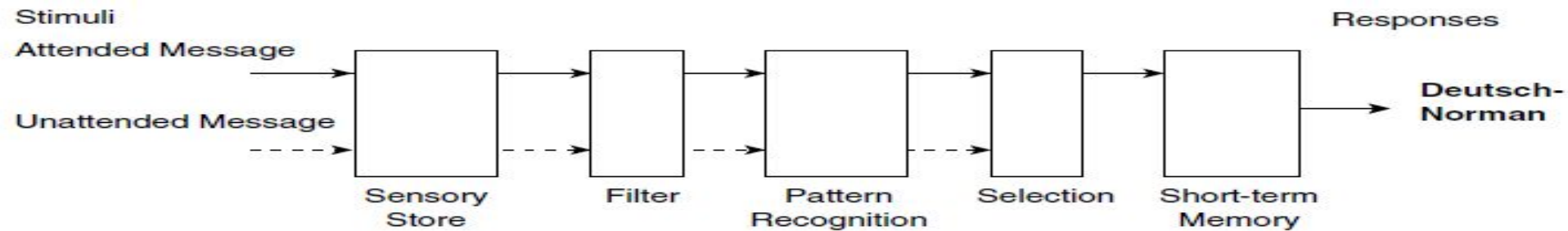
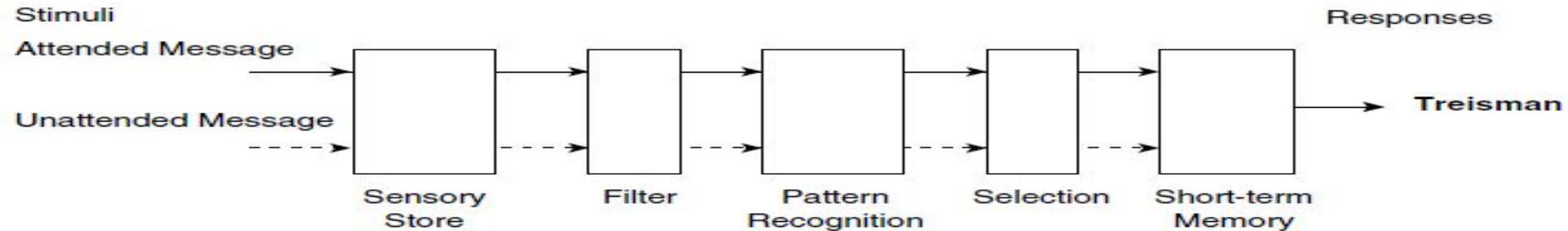
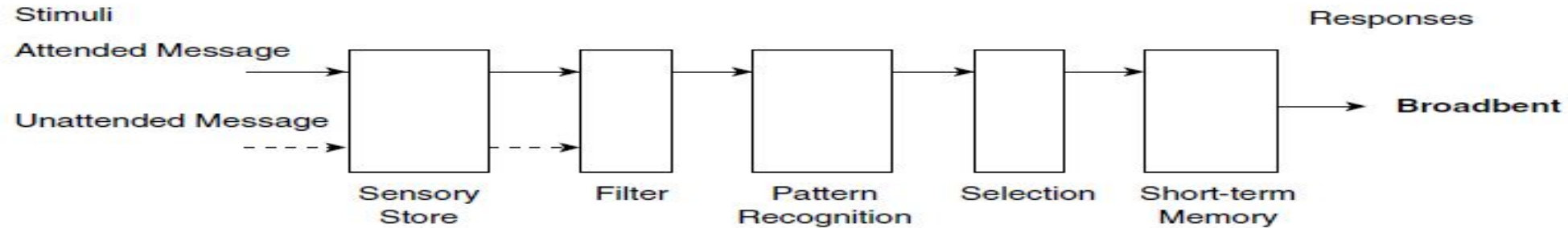
Theories of Attention

- **The Deutsch-Norman Memory Selection Model**

- The first stages of processing in this model are the same as those of the other models.
- Before being passed into short-term memory, it goes through a secondary selection mechanism..... selects information on the basis of semantic characteristics or message content.
- The selected items end up in short-term memory and awareness.



Theories of Attention



Information processing models of selective attention. The solid arrows indicate information from the attended message. The dashed arrows indicate information from the unattended message

Theories of Attention

- **The Multimode Model of Attention**

- A more current view is that selection can occur early or late.
- Michael Posner and Charles Snyder (1975) advocate the idea of a “moveable filter,” one that can operate at various points in processing according to the observer’s needs.
- Data from another study support what seemed to some to be the case all along, that selection can be based on physical or semantic properties
- Semantic selection imposes a greater cost, that is, it requires greater attentional resources than physical selection.

Theories of Attention

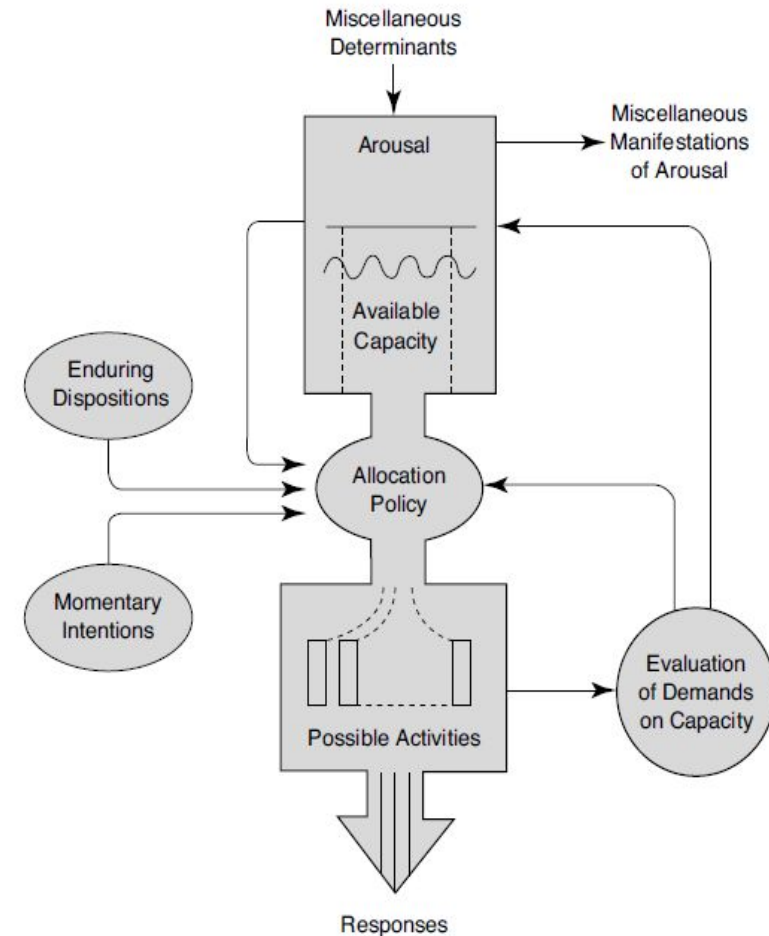
- **The Multimode Model of Attention**

- The model that arises from the view that selection can be based on multiple modes, the physical and the semantic, is called the **multimode model of attention**.
- There is also neurophysiological evidence to support the multimode model.
- Attentional selection is more flexible than was first assumed.
- It can happen early or late, and can be based on more than just a single mode.

Theories of Attention

- **Kahneman's Capacity Model of Attention**

- Capacity models describe attention as a resource.
- We can think of this resource as the amount of mental effort or energy required to perform a task.
- Like many other resources, attention exists in limited supply. At any given time, we only have so much of it that is available for use
- Capacity models are really models that describe the division, rather than the selection, of attention.



- **Arousal** refers to physiological activation and is reflected in values such as cardiac and respiratory rates.

Theories of Attention

- *Evaluating the Capacity Model of Attention*
 - Stephen Reed, in his book *Cognition* (2000) noted that capacity models such as Kahneman's are not designed to replace selection models, but rather to supplement them.
 - Capacity models give us a good overview of the many influences and interdependencies that are involved in the allocating of attention.
 - Selection models provide us with a better feel for the discrete stages of information flow during attentional processing.

Cognitive Science Influences AI

- AI has been profoundly influenced by the **concept of attention**, a cornerstone of human cognitive processes.
- This development set the stage for the incorporation of attention mechanisms in AI.
- **Attention mechanisms** allow the AI to selectively focus on specific input data segments, mimicking how human attention operates.
- **Example 1:-** In image recognition, attention mechanisms allow for a more focused analysis of specific image regions, leading to higher accuracy.
- **Example 2:-** In NLP attention helps construct more coherent and contextually appropriate responses.

Cognitive Science Influences AI

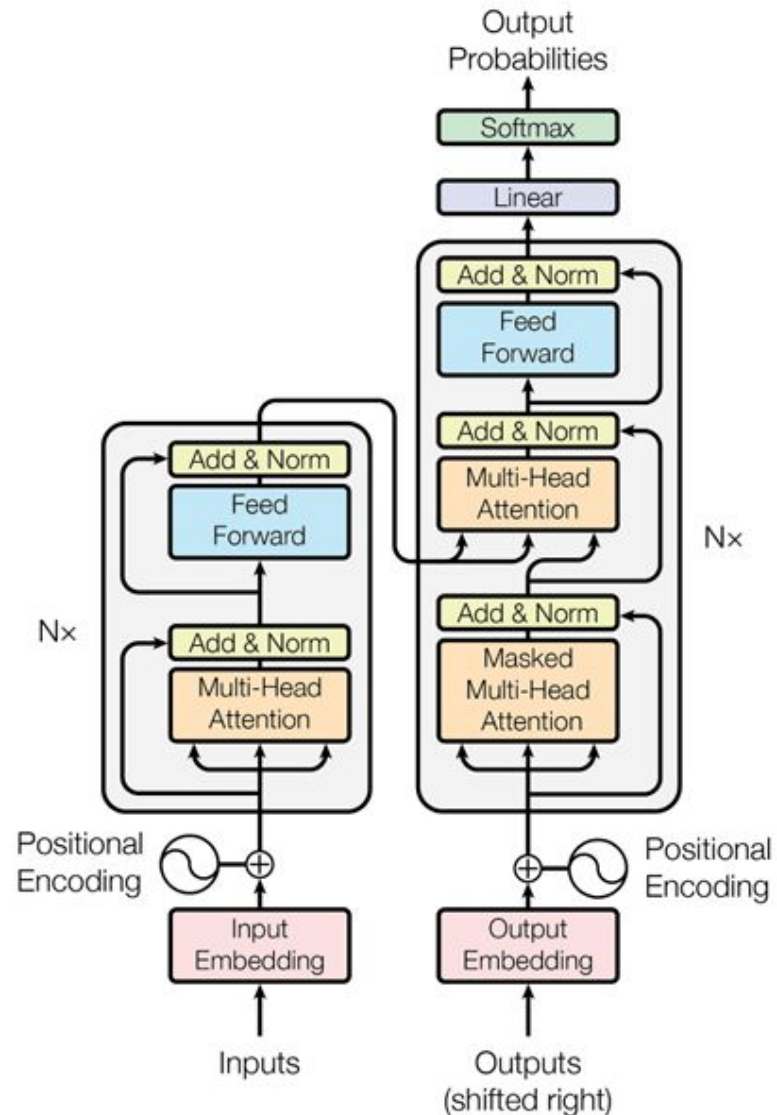
- **Autonomous Driving and Pedestrian Detection:** In a self-driving car's visual system, the camera captures a massive amount of data: the sky, the sidewalk, buildings, and the road.
 - **Without Attention:** The AI might treat every pixel with equal importance.
 - **With Attention:** The mechanism allows the car's AI to "attend" specifically to moving objects or high-risk areas
 - A child standing near the curb or
 - A brake light flashing three cars ahead—even if those objects are small in the overall frame.

Cognitive Science Influences AI

- Today's most advanced AI models, such as OpenAI's GPT series, Claude, and Gemini heavily leverage attention mechanisms.
- These models employ **transformer architecture**, fundamentally built around the concept of attention, enabling an unprecedented level of sophistication in language processing.
- It uses **Multi-Head Attention**, which means it runs the attention process multiple times in parallel.
 - One might focus on grammar (linking a verb to its subject).
 - The other might focus on pronouns (linking "he/She" to "Biruk").
 - The third might focus on intent or emotion.
- By combining these different perspectives, the model builds a multi-dimensional understanding of the sentence.

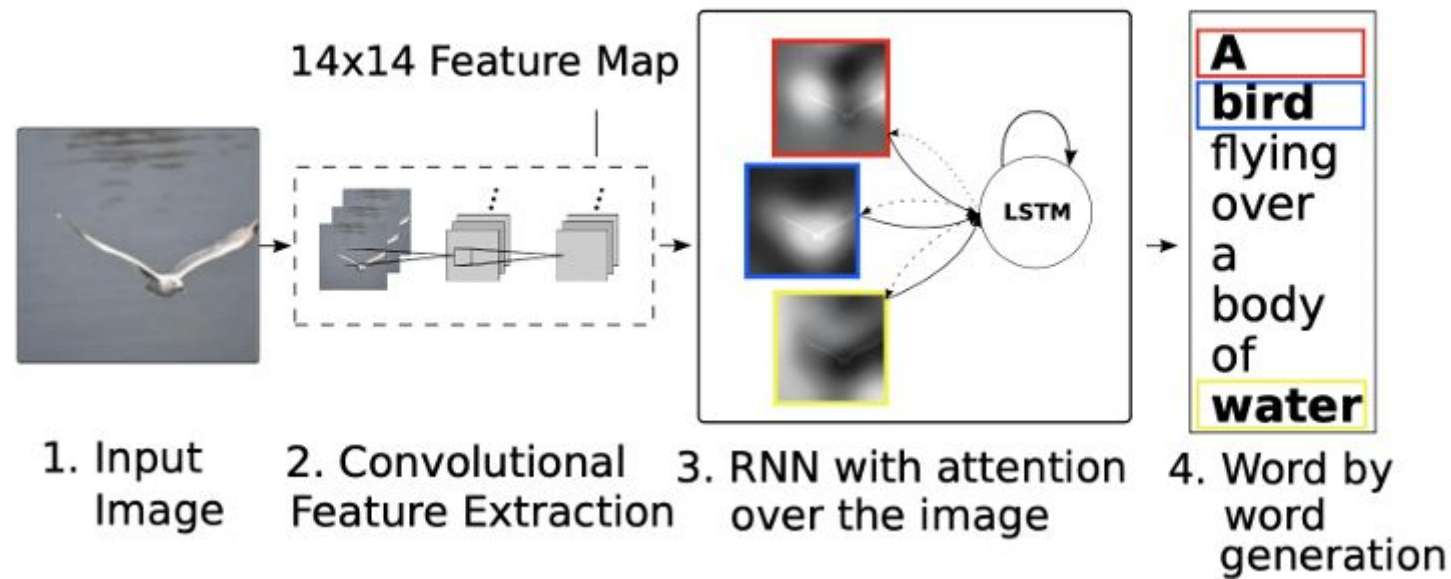
Cognitive Science Influences AI

- **Encoder**, on the left half of the transformer architecture, is to map an input sequence to a sequence of continuous representations, which is then fed into a decoder.
- **Decoder**, on the right half of the architecture, receives the output of the encoder together with the decoder output at the previous time step to generate an output sequence.
- While the encoder is designed to attend to all words in the input sequence regardless of their position in the sequence, the decoder is modified **to attend only** to the preceding words.
- The attention mechanism connects the encoder and decoder



The encoder-decoder structure of the transformer architecture.

Cognitive Science Influences AI



Reading assignment:
The transformer
architecture

Neural Image Caption Generation with Visual Attention

The Cognitive Approach: Memory, Imagery, and Problem Solving

- Focus on three other prominent cognitive processes: memory, imagery, and problem solving.
- **Memory**, very generally defined, is the capacity to retain information over time.
- Memory allows us to store our past experiences and draw on them.
- Any discussion of memory must begin with the notion that there is no single type of memory.
- Cognitive research in this area has led to a rather consistent finding: the existence of functionally distinct memory systems.

The Cognitive Approach: Memory

- **Sensory Memory**

- A repository for incoming sensory information
- Maintain the representation of a stimulus long enough so that it can be recognized.
- There are different forms of sensory memory, there is one for each of the five senses.
 - Iconic memory is a visual sensory memory, stores visual representations.
 - Echoic memory is an auditory sensory store, auditory ones representations .

The Cognitive Approach: Memory

- Working Memory
 - Is also sometimes known as short-term memory
 - The duration of items residing in working memory is, as we will see, much longer than that of items residing in sensory memory.
 - Working memory retains less information; its capacity is unlike the unlimited capacity of the visual icon.
 - Coding in working memory can be acoustic, semantic, or visual.
 - It is a place where data (the parts) can be temporarily stored so that they can be operated on by cognitive processes (the tools).

The Cognitive Approach: Memory

- Brown (1958) and Peterson and Peterson (1959) were able to determine that the duration of short-term memory was about 18 seconds.
- How much can working memory hold?
- On average, individuals can retain about seven items, give or take two, commonly denoted as 7 ± 2 , this limit has come to be called the “magical number seven” (Miller, 1956).
- But what exactly is an “item”? Is it a number? A letter? A word?
- To understand this, let's start with a demonstration.

The Cognitive Approach: Memory

N F L E B C I R A M T V

The Cognitive Approach: Memory

- You may at this point notice that the letters cluster in groups of three—each group is a popular acronym. “NFL” stands for “National Football League,” “CBS” for “Ethiopia Board Casting Corporation,” and so on.
- If you had noticed these groupings, it would have been easy for you to remember all the letters.
- This is because you would then be remembering only four groups of three letters and not twelve individual letters.
- A group of meaningful items is called a chunk.
- The grouping of items into a single meaningful whole in short-term memory is known as **chunking**.
- We can increase the total amount of information that is contained in short-term memory by chunking: the bigger the chunk, the greater the capacity.

How is information coded or represented in working memory?

The Cognitive Approach: Memory

- How is information coded or represented in working memory?
- Can be coded in a variety of formats.
- Conrad (1964) presented letter strings to participants and then asked them to perform immediate recall.
- He found that although the letters had been presented visually, the pattern of errors was based on the sounds of the letters. Participants would mistake an “A” for a “K,” for example, or an “E” for a “P.”
- This suggests that the letters were being converted to an **acoustic code**, one based on the sounds of the items.
- Words were represented not acoustically, but instead on the basis of their inherent meanings—the code was a **semantic code**.
- A **visual code** is one that preserves spatial characteristics.

The Cognitive Approach: Memory

- Long term memory
 - Individuals are capable of remembering information for longer than just a few seconds.
 - Many of us can recall quite vividly events from our childhoods or facts learned decades ago in school.
 - We must therefore have another memory system, one that is capable of storing data for longer periods of time.
 - long-term memory has several distinct types.
 - *Procedural memory* holds procedural knowledge.
 - Is memory for skill, is demonstrated by doing, and arises without conscious recall.
 - Procedural memory also sometimes goes by the name of implicit memory.
 - *Declarative memory* contains declarative knowledge.
 - It is memory for facts and events, is demonstrated by speaking, and arises with conscious recall.

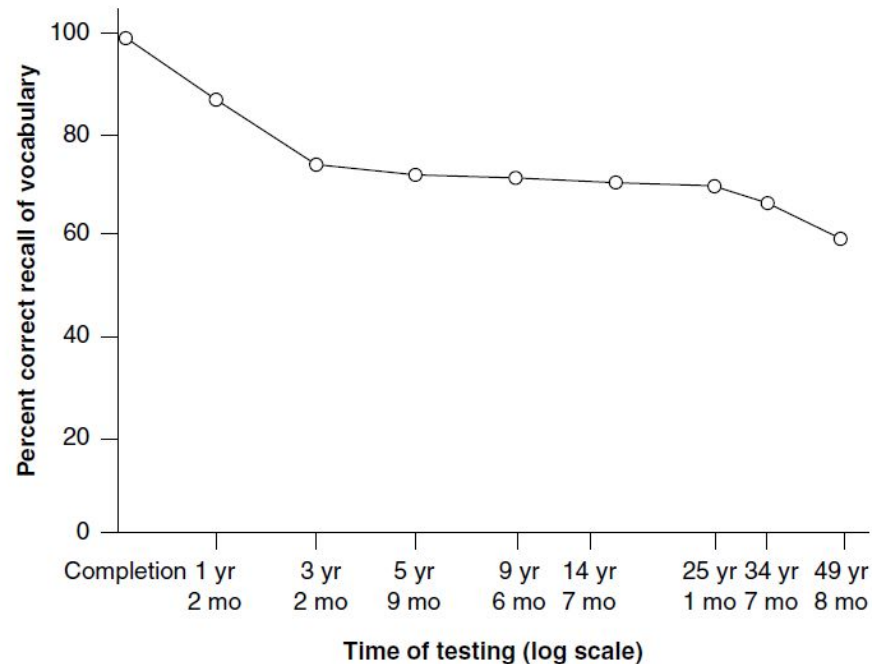
The Cognitive Approach: Memory

- There are two types of declarative memory.
 - **Semantic memory** is knowledge of facts and general knowledge of the sort learned in school.
 - **Episodic memory** contains episodes, or personally experienced events, for example, what you did on your birthday last year.
- We are usually consciously aware of declarative information, which is sometimes referred to as **explicit memory**.
- A study by Bahrick (1984) shows us **how long information in semantic memory lasts**.

The Cognitive Approach: Memory

- Bahrick studied a group of people who studied Spanish early in life, usually during the high school or college years.
- Three distinct memory stages were discovered
 - Stretching from 3 to 6 years after the classes, there was an initial rather dramatic loss of information.
 - Stretching from 6 to up to 30 years after initial learning, there was no further loss—the amounts remembered stayed relatively constant.
 - Stretching from 30 to 35 years later, there was another loss of information, but only a slight one.

The Cognitive Approach: Memory



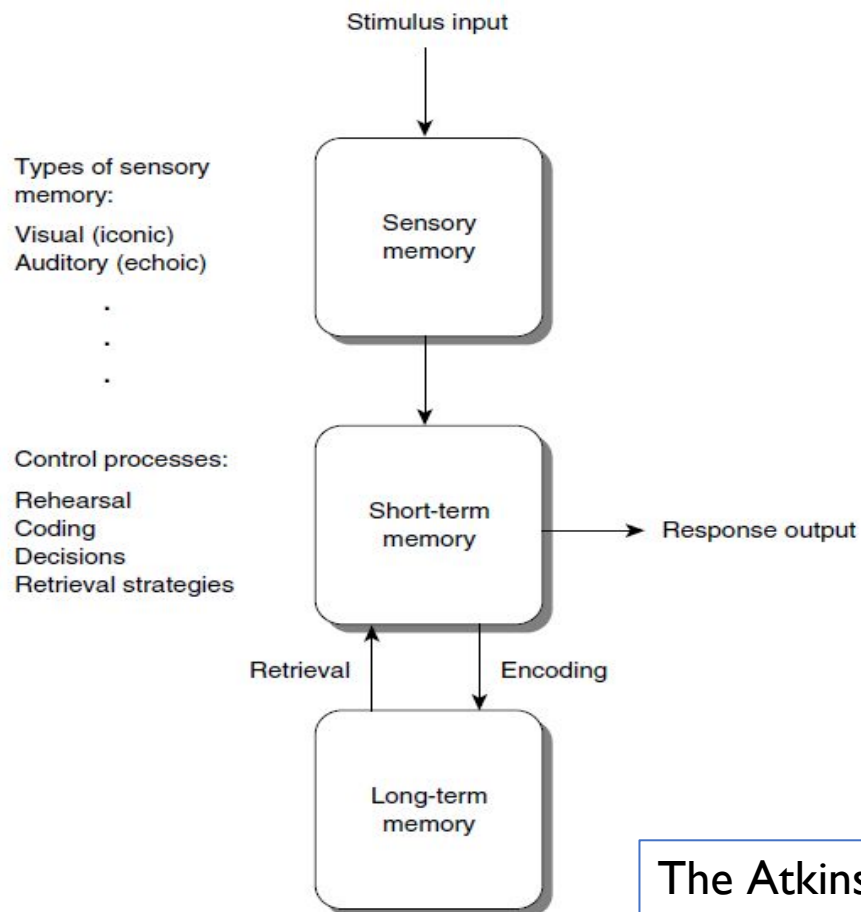
- Information in long-term storage, if not used or rehearsed, decays fairly rapidly over the first few years.
- There is some residual amount of learned information that remains intact over a long period of time.
- There appears to be a slight additional loss of information that occurs later in life.
- How much can be retained in semantic long-term memory?
- It has been proposed that we remember virtually everything we've ever experienced in our entire lives, but simply have difficulty recalling it.

Can you compare this with AI Machines?

The Cognitive Approach: Memory

- **Memory Models**

- Information processing theories specify interactions between all the different memory types we've seen so far.

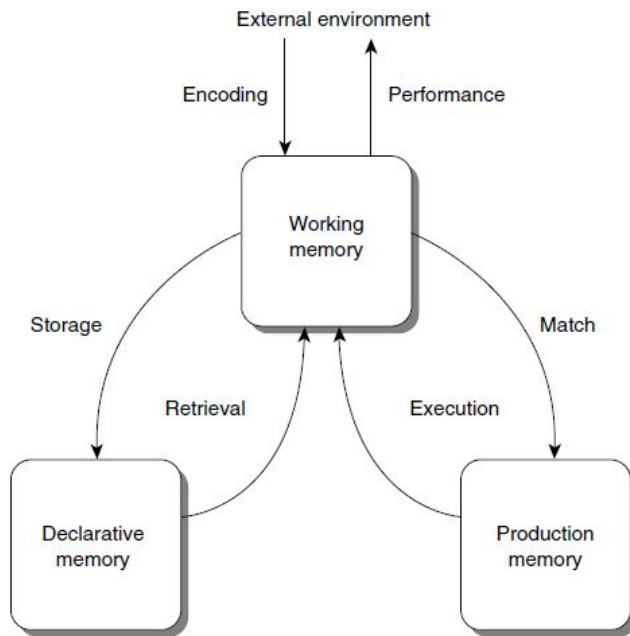


- The **modal model** of memory was **the first model** to provide a general overview of how information is processed in each of the different memory types (Atkinson & Shiffrin, 1971).
- **Encoding** is the name of the process by which information is taken into long-term memory and converted into a usable form.
- Once in long-term memory, information may not be used immediately. We say that the information is in a state of **storage**.
- Was developed very early in the **course of cognitive psychology** and fails to specify many of the nuances of memory structure and function.

The Atkinson & Shiffrin (1968) modal memory model

The Cognitive Approach: Memory

- **The ACT* Model** (stands for Adaptive Control of Thought)
 - John Anderson (1983, 1990) proposes a global model of memory function that is similar to the modal model.



- Has three components: working memory, declarative memory, and production memory.
- Declarative memory is equivalent to explicit memory and production memory is equivalent to implicit memory.
- Encoding-match-execution performance loop is activated when one is carrying out any learned skill, whether driving, skiing, or playing the piano.
- *Example: Now imagine that you are answering a factual multiple-choice question on a college exam.*

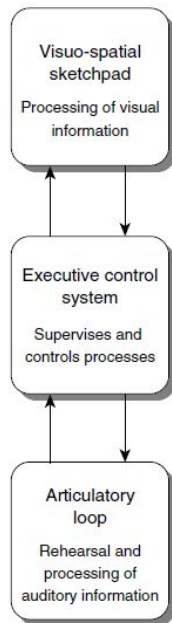
The Cognitive Approach: Memory

- The strength of Anderson's model lies in its description of separate processing loops for implicit and explicit memories.
- Notice that there are no arrows that connect production memory and declarative memory--- these are considered two entirely distinct and independent memory stores.
- Anderson postulates an additional component of this model that represents how propositional information in declarative memory is organized.

The Cognitive Approach: Memory

- **The Working Memory Model**

- Baddeley (1986, 1992) has formulated a detailed model for the components and processes of working memory..... composed of three separate units.



- Primary unit --- the **executive control system**.
 - Initiate and control ongoing processes
- The second major component is the **articulatory loop**, sometimes called the phonological loop.
 - Speech and sound-related information are rehearsed
- Third unit, called the **visuospatial sketchpad**, is specialized for the processing of visual information.
- Both the loop and the sketchpad are “slave systems” to the central executive.
- Recent studies have shown that there are specific brain regions that correspond to the different components of the Baddeley model

The Cognitive Approach: Imagery

- **Visual Imagery**

- How many windows are there in your home?
- A **visual image** is a mental representation of an object or scene that preserves metric spatial information.
- Visual images are therefore isomorphic to their referents.
- They preserve spatial characteristics of the objects they represent.
- A defining hallmark of imagery is that, unlike perception, it is not driven by sensory input from the outside world.
- Imagery can be driven by information coming from memory, an internal information source.

The Cognitive Approach: Imagery

- Numerous studies have shown that there are many similarities between perceptual processes and imagery.
- Both processes seem to draw on the same underlying neural machinery.

“Goldenberg et al. (1988) showed that when participants are asked to answer questions that require the use of visual imagery, there is a greater increase in vision-related brain activity, measured as increased blood flow to the visual cortex, than in participants who are given questions that do not require the use of imagery.”

- It is important to note that “imagery” in a more general sense refers to the mental representation of something and need not be restricted to the visual domain.

The Cognitive Approach: Memory, Imagery, and Problem Solving

- **The Kosslyn and Schwartz Theory of Visual Imagery**

- An early and influential theory of imagery..... functional theory, it describes what the brain does with images.
- *Image Structures*
 - Kosslyn posits two kinds of image structures
 - A **surface representation** is a quasi-pictorial representation that occurs in a spatial medium.
 - He describes the medium as a visual buffer.
 - **Deep representations** constitute the second type of image structure.
 - They consist of information in long-term memory that is used to generate the surface representations.

The Cognitive Approach: Memory, Imagery, and Problem Solving

- **The Kosslyn and Schwartz Theory of Visual Imagery**

- Kosslyn describes two classes.
- **Literal encodings** contain lists of coordinates that detail the placement of points in the surface matrix such that the represented object is depicted..... some evidence suggests **they originate in the right hemisphere**.
- **Propositional encodings** are abstract, language-like representations, similar to declarative statements.
 - They contain information about an object's parts, the locations of these parts, and their sizes... they are believed to lie in the **left hemisphere**.

The Cognitive Approach: Memory, Imagery, and Problem Solving

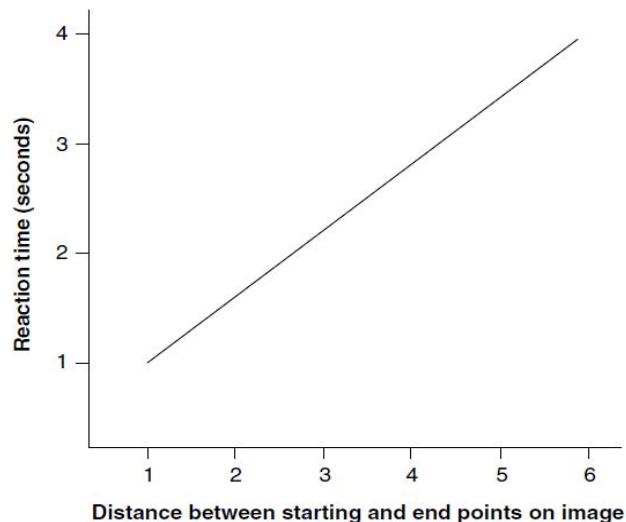
- *Image Processes*

- Three types of image processes.
 - Image generation occurs when the encodings in long-term memory are used to form an image in the visual buffer.
 - One question concerning image generation is whether images are formed all at once or bit by bit.
 - The evidence suggests that they are formed bit by bit (Kosslyn, 1980).
 - Images that have more parts take longer to create.... the same is true for images that have more detail.

The Cognitive Approach: Memory, Imagery, and Problem Solving

- *Image Processes*

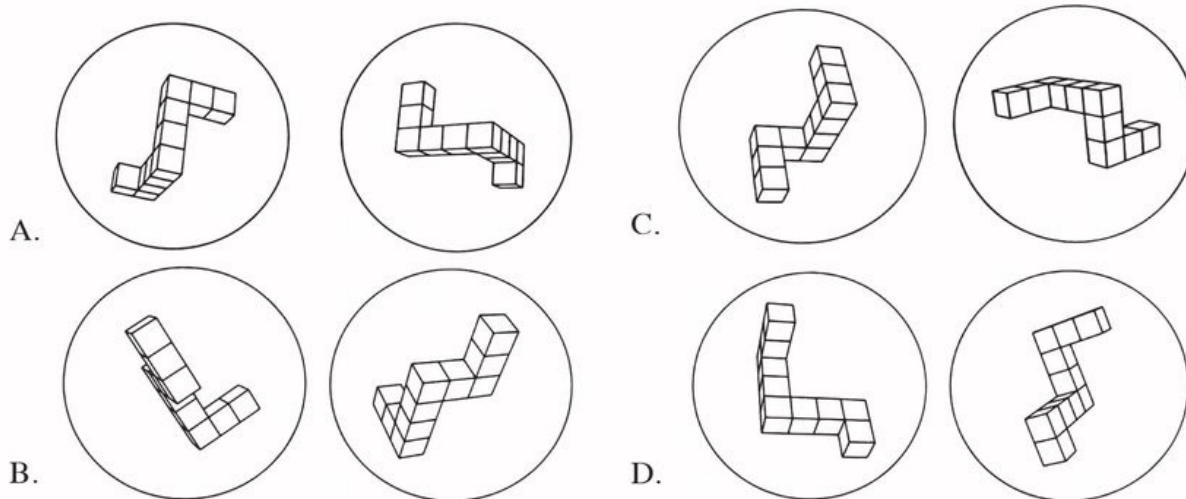
- Three types of image processes.
 - **Image inspection** occurs when we are asked some question about an image
 - He asked participants to imagine a large object next to a small one.
 - Subjects were asked to imagine an elephant standing next to a rabbit. They were then asked to respond to questions, for example: “Does the rabbit have a tail?”
 - Participants took longer to respond to questions about the rabbit. His explanation was that the rabbit was relatively small in the image and therefore difficult to resolve.



Idealized results from a visual image study showing time needed to mentally scan from one point of an image to another as a function of the distance between objects

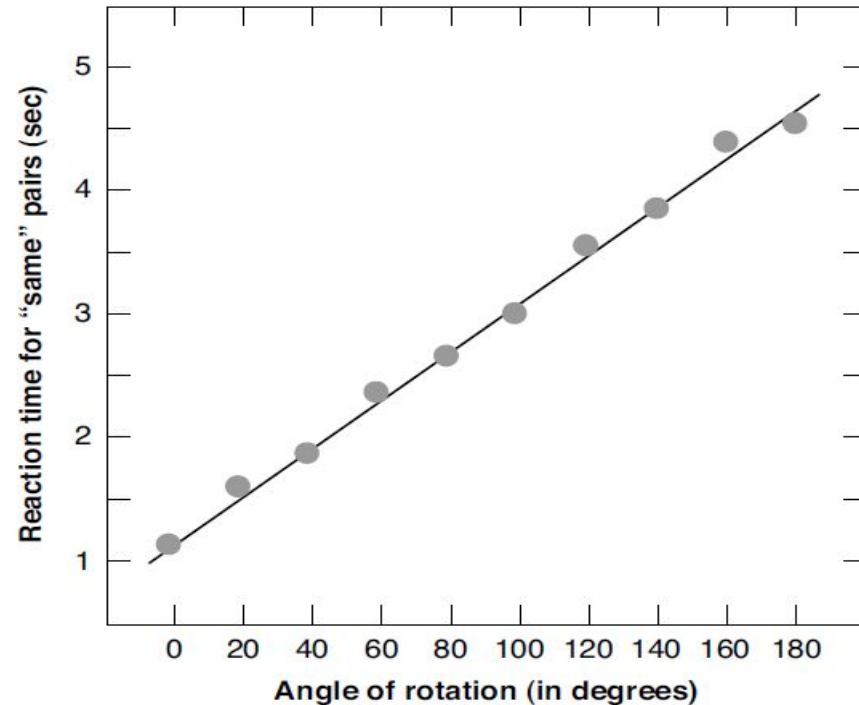
The Cognitive Approach: Memory, Imagery, and Problem Solving

- **Image transformation** refers to an operation performed on an image.
 - Kosslyn describes two transformation modes—shift transformations and blink transformations.
 - Most image transformations are shift transformations, in which the image is altered incrementally.
 - Image passes through intermediate stages along a trajectory of size, orientation, or location.
 - **Blink transformations** alter the entire image at once and can be used to re-form a complete image after it has decayed in the buffer.
 - A classic experiment in cognitive psychology calls attention to an example of a shift transformation. Shepard and Metzler (1971) presented participants with drawings of pairs of three-dimensional objects



Stimuli from an early mental rotation experiment. Participants were asked to judge whether the two patterns were rotated versions of each other

The Cognitive Approach: Memory, Imagery, and Problem Solving



- This graph illustrates a famous discovery in cognitive psychology called Mental Rotation.
- At 0° the reaction time is very fast (~1 second) while at 180° the reaction time is much slower (~4.5 seconds).
- This graph proves that humans don't just instantly recognize an upside-down object. **Instead, our brains perform a physical-like simulation**

Response time plotted as a function of the angular difference between the two patterns suggests a mental rotation

The Cognitive Approach: Memory, Imagery, and Problem Solving

- *Evaluating the Kosslyn and Schwartz Theory*

- Pylyshyn (1981) provides several critiques of the theory.
- He says there are too many free parameters, that is, functions or assumptions that are uncontrolled and that can affect the results of any experiment designed to test the theory.
- A second critique concerns **cognitive penetration**, in which one's knowledge, beliefs, goals, or other cognitive states alter performance of a mental task.

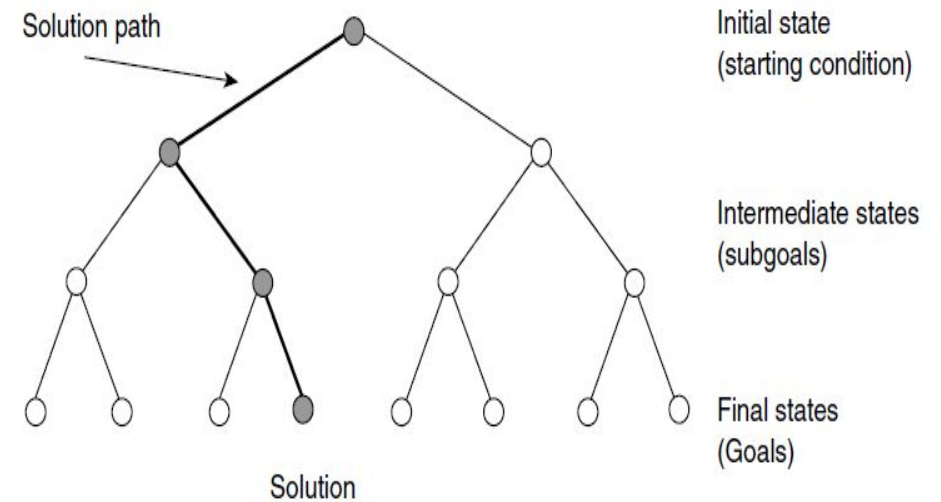
The Cognitive Approach: Problem Solving

- **Problem Solving**

- Problem solving is the transformation of a given situation into a desired situation or goal (Hayes, 1989).
- Anderson (1980, 1985) lists four characteristics of problem solving:
 - **Goal directedness.** Problem solving behavior is directed toward the attainment of a goal, which is the desired end point or solution to the problem.
 - **Sequence of operations.** problem solving behavior involves a sequence of steps.
 - **Cognitive operations.** the application of some process to the problem solving situation that transforms the situation.
 - **The setting of subgoals.** Each step in the problem-solving sequence produces a subgoal

The Cognitive Approach: Memory, Imagery, and Problem Solving

- A **problem space** may be thought of as the different situations or states that can exist in a problem.
- Three important states.
 - The **initial state** is what the problem is like at the outset.
 - **Intermediate states** are what results after some action is applied to the problem.
 - The **goal state** is a desired situation and can include the final situation or configuration of the problem after a solution has been obtained.
- Problem spaces are usually represented in a diagram, called a **solution tree**



A representation of a problem space showing the solution path

The Cognitive Approach: Problem Solving

- One approach to problem solving is to try all possible paths represented in the solution tree.... is a **brute force method.. computationally intensive**
- **Heuristic** is an informal “rule of thumb” method of problem solving that **does not guarantee a solution** but is faster and easier to use than a systematic search.
- The most effectual heuristic in problem solving is the **means-end analysis**.
 - Problem is solved via **successive determinations of the difference between** the existing state and the goal or sub-goal state.... focuses on reducing the ‘distance’ between where you are now and where you want to be.
- We all use means-end problem solving in our everyday life.

The Cognitive Approach: Problem Solving

- **The General Problem Solver Model (GPS)**

- Alan Newell and Herb Simon were early pioneers in problem solving research.
- They developed GPS – by applying means-end analysis.
- A key component of the model was its use of production rules
- The GPS program has been applied to several well-known logic puzzles, including the Tower of Hanoi problem, the Missionary-Cannibal problem, the three coins puzzle, the water jug task, and others.

The Cognitive Approach: Problem Solving

- *Evaluating the General Problem Solver Model*
- GPS has its limitations... it diverges from the strategies human beings use to solve problems.
- With some problem solving, human beings are apt to use another heuristic, called the hill-climbing strategy... it involves taking actions that always bring one closer to the goal.

Cognitive approach vs AI in memory, imagery and problem solving?

What are the biological bottlenecks that prevent us from being perfect information processors?

Cognitive approach vs AI in memory, imagery and problem solving?

- Limitations of cognitive approach
 - Prone to errors (e.g., false memories, interference), capacity constraints (e.g., STM holds $\sim 7 \pm 2$ items), and degradation over time
 - Fuzzy and fleeting; mental images lack the precision of real visuals and degrade quickly without focus
 - Slow, error-prone, and biased (e.g., confirmation bias). Cognitive load limits complex multitasking or exhaustive computation.

Cognitive approach vs AI in memory, imagery and problem solving?

- Artificial Memory

- **Mechanism:** AI memory relies on data storage systems (e.g., databases, RAM, neural network weights).
 - Retrieval is deterministic or probabilistic, depending on the model (e.g., exact lookup vs. generative recall in LLMs).
- **Strengths:** Vast capacity, precise recall of stored information, and no natural decay unless data is corrupted or overwritten.
 - AI can process and index massive datasets humans couldn't handle.
- **Limitations:** Lacks emotional or contextual nuance unless explicitly programmed.

Cognitive approach vs AI in memory, imagery and problem solving?

- Artificial Imagery
 - Mechanism: AI generates imagery using models like GANs (Generative Adversarial Networks) or diffusion models.
 - It learns patterns from vast image datasets and synthesizes new visuals based on prompts or parameters.
 - Strengths: Produces detailed, high-resolution outputs with consistency and speed.
 - AI can create visuals beyond human skill (e.g., photorealistic scenes) and iterate endlessly.
 - Limitations: Reliant on training data—AI struggles with truly novel concepts outside its dataset.

Cognitive approach vs AI in memory, imagery and problem solving?

- Art
 - October 25, 2018 was landmark in AI history
 - AI-created painting called “Portrait of Edmond de Belamy” was sold at Christie’s in New York for over 300,000 pound



- Initially the AI system provided with 15,000 portraits.
- A generative adversary network was used
- The generator created new AI portrait based on the dataset then the discriminator tried to detect difference between that portrait and a human-created portrait
- The process repeated so many times until the discriminator decided a given AI portrait is painted by a human being

Cognitive approach vs AI in memory, imagery and problem solving?

- **Artificial Problem-Solving**
 - **Mechanism:** AI solves problems via algorithms (e.g., search, optimization) or machine learning (e.g., pattern recognition).
 - Techniques range from brute-force searches (e.g., chess engines) to probabilistic inference (e.g., in LLMs).
 - **Strengths:** Lightning-fast at well-defined problems (e.g., solving equations, optimizing routes).
 - AI can process vast datasets and identify patterns humans miss.
 - **Limitations:** Struggles with ambiguity or problems lacking clear rules/data. AI lacks true insight or ethical judgment unless programmed.

Thank You